sent the Mangalyaan spacecraft to Mars when it was at a distance of close to 99 million km from Earth. However, to do so, it had to traverse 660 million km because of the orbit-raising manoeuvres it needed to make.

This expertise will allow ISRO to load a lot more instruments onto Aditya L1 to comprehensively study critical solar parameters that impact us on Earth. And why is that important? Because, as we all know, the Sun wields great influence on Earth with radiation, heat and continuous flow of particles called solar winds, in addition to magnetic fields that affect the entire solar system. Occasionally, there are explosive and eruptive events like Coronal Mass Ejections (CME) or solar flares that, among other things, trigger a magnetic storm near Earth and interfere with the functioning of space-based and terrestrial telecommunication systems. As Kiran Kumar, former ISRO chairman, says, "It is important that nations study such disruptive phenomena of the Sun, make sense of them and thereby enable early warning signals to Earth to minimise any damage these may cause." Kumar also points out how Aditya's orbital path will give the spacecraft an uninterrupted view of the Sun, enabling its instruments to study CME and solar coronal dynamics.

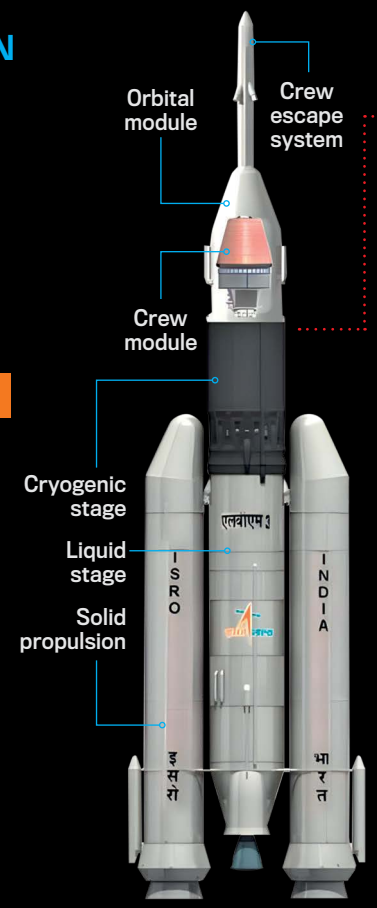
To allow Indian scientists to study such phenomena closely, Aditya is bristling with seven scientific payloads put together not just by ISRO but also by other major Indian scientific institutions. The Bengaluru-based Indian Institute of Astrophysics, for instance, has developed the Visible Emission Line Coronagraph (VELC) to study both the solar corona and CME dynamics. The Inter-University Centre for Astronomy and Astrophysics in Pune has built the Solar Ultraviolet Imaging Telescope (SUIT), which will capture images of the solar photosphere and chromosphere and measure its irradiance to throw new light on its composition. Ahmedabad's Physical Research Laboratory (PRL) has put together the Aditya Solarwind Particle Experiment (ASPEX), which along with the Plasma Analyser Package for Aditya (PAPA) built by the Space Physics Laboratory in Thiruvananthapuram, will study the solar wind and energetic ions, as well as their energy distribution. Two payloads—the Solar Low Energy X-ray Spectrometer (SoLEXS) and the High Energy L1 Orbiting X-ray Spectrometer (HEL1OS) to study the X-ray flares from the Sun—and another, the Magnetometer, to measure interplanetary magnetic fields, are built by two of ISRO's Bengaluru-based centres, namely the URSC and the Laboratory for Electro Optics Systems (LEOS), respectively. Aditya, in fact, is a good example of the phenomenal collaborative effort of scientific institutions that gives ISRO the cutting edge in whatever it undertakes.

Meanwhile, the most significant transformation ISRO is going through in its 54 years of existence is on account of the radical policy changes the Modi government has introduced in India's space sector. Prior to 2020, ISRO planned, designed and built all of India's spacecraft, be it rocket launchers, satellites or planetary orbiters. To its credit, the space agency engaged the private sector, but mainly to manufacture its sub-systems and assist it in assembling the spacecraft. In May 2020, however, in the midst of the Covid-19 pandemic, Prime Minister Narendra Modi threw open the gates of the space sector to comprehensive private participation. This included permitting private sector players to build and assemble their own rockets and satellites

GAGANYAAN
OUR FIRST HUMAN SPACE FLIGHT

THE MISSION

- Demonstrate India's human spaceflight capability
- Launch a crew of three members to an orbit of 400 km for a three-day mission
- Bring them back safely to Earth, by landing in Indian sea waters
- Planned launch: 2025-26



PRIVATE INVESTMENT WILL BE IMPORTANT IF INDIA'S SPACE SECTOR HAS TO EXPAND RAPIDLY. IT CANNOT DEPEND ON GOVT FUNDING ALONE